



## Microcontrollers

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## Supply Voltage Supervisor (SVS)

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The SVS is used to monitor the  $A_{VCC}$  supply voltage or an external voltage.

The SVS can be configured to set a flag or generate a POR reset when the supply voltage or external voltage drops below a user-selected threshold.

The SVS features include:

- AVCC monitoring
- Selectable generation of POR
- Output of SVS comparator accessible by software
- Low-voltage condition latched and accessible by software
- 14 selectable threshold levels
- External channel to monitor external voltage

# Supply Voltage Supervisor (SVS)

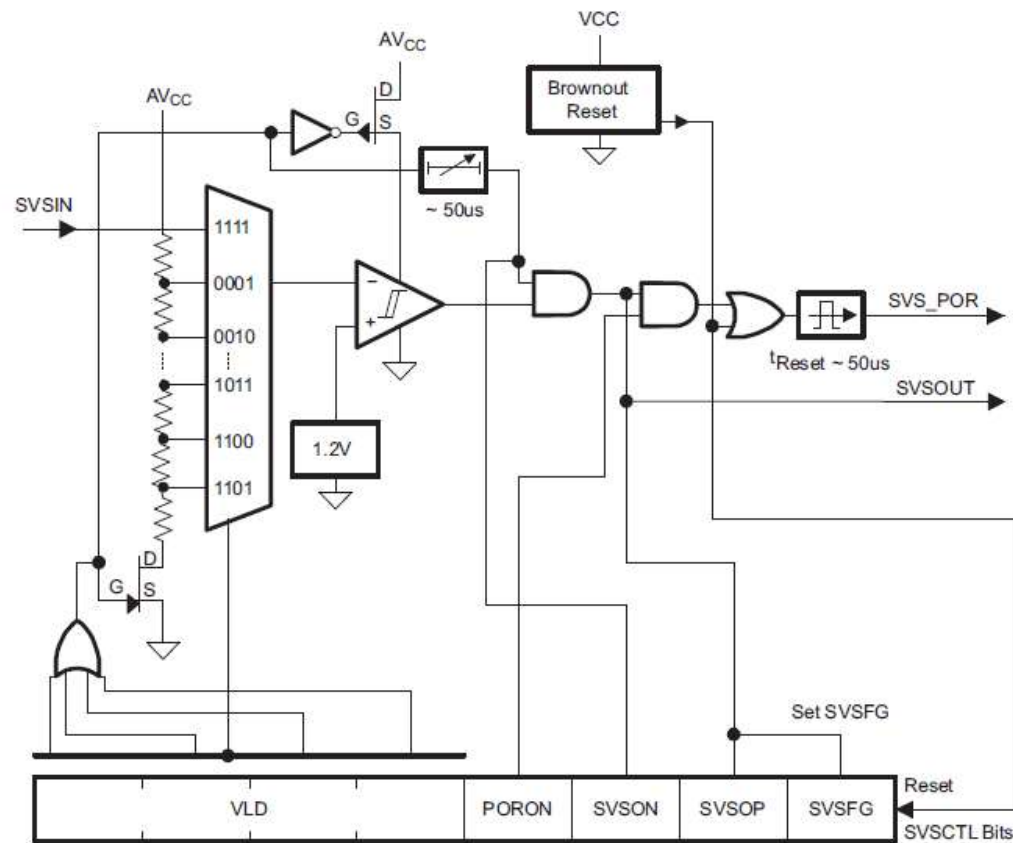


Figure 9-1. SVS Block Diagram

## Supply Voltage Supervisor (SVS)

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### SVS Operation

The SVS detects if the  $A_{VCC}$  voltage drops below a **selectable level**.

It can be configured to provide a POR or set a flag, when a low-voltage condition occurs.

The SVS is disabled after a brownout reset to reduce power

### Configuring the SVS

The VLDx bits are used to enable/disable the SVS and select one of 14 threshold levels ( $V(SVS\_IT-)$ ) for comparison with AVCC.

The SVS is off when  $VLDx = 0$  and on when  $VLDx > 0$ .

The SVSON bit does not turn on the SVS.

It reflects the on/off state of the SVS and can be used to determine when the SVS is on.

When  $VLDx = 1111$ , the external SVSIN channel is selected. The voltage

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When  $VLDx = 1111$ , the external SVSIN channel is selected.

The voltage on SVSIN is compared to an internal level of approximately 1.25 V.

### SVS Comparator Operation

A low-voltage condition exists when  $A_{VCC}$  drops below the **selected threshold** or when the **external voltage** drops below its **1.25-V threshold**.

Any low-voltage condition sets the SVSFG bit.

The PORON bit enables or disables the device-reset function of the SVS. If  $PORON = 1$ , a POR is generated when SVSFG is set. If  $PORON = 0$ , a low-voltage condition sets SVSFG, but does not generate a POR.

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The SVSFG bit is latched.

This allows user software to determine if a low-voltage condition occurred previously.

The SVSFG bit **must be reset** by user **software**.

If the low-voltage condition is still present when SVSFG is reset, it will be immediately set again by the SVS.

## Supply Voltage Supervisor (SVS)

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### Changing the VLDx Bits

When the VLDx bits are changed from zero to any non-zero value there is an automatic settling delay  $t_d(\text{SVSon})$  implemented that allows the SVS circuitry to settle.

The  $t_d(\text{SVSon})$  delay is approximately 50  $\mu\text{s}$ .

During this delay, the SVS will not flag a low-voltage condition or reset the device, and the SVSON bit is cleared.

Software can test the SVSON bit to determine when the delay has elapsed and the SVS is monitoring the voltage properly.

Writing to SVSCTL while SVSON = 0 will abort the SVS automatic settling delay,  $t_d(\text{SVSon})$ , and switch the SVS to active mode immediately.

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When the VLDx bits are changed from any non-zero value to any other non-zero value the circuitry requires the time  $t_{\text{settle}}$  to settle.

The settling time  $t_{\text{settle}}$  is a maximum of  $\sim 12 \mu\text{s}$  (device-specific)

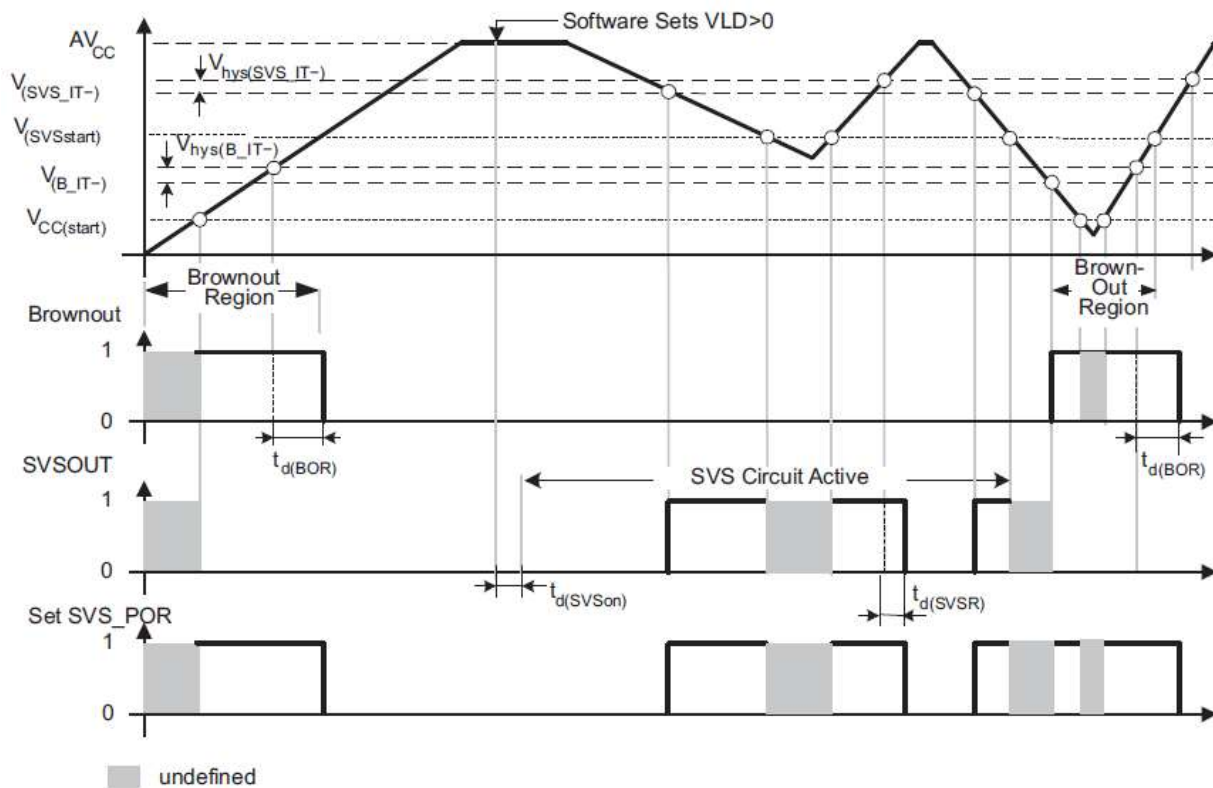
The recommended flow to switch between levels is shown in the following code.

```
MOV.B #080h,&SVSCTL      ; Enable SVS for the first time:
                           ; Level 2.8V, do not cause POR
                           ; ...
                           ; Change SVS level
MOV.B #000h,&SVSCTL      ; Temporarily disable SVS
MOV.B #018h,&SVSCTL      ; Level 1.9V, cause POR
                           ; ...
```



## SVS Operating Range

Each SVS level has hysteresis to reduce sensitivity to small supply voltage changes when AVCC is close to the threshold.



## SVS Operating Range

Figure 9-3. SVSCTL Register

| 7                   | 6                   | 5                   | 4                   | 3                   | 2                  | 1                  | 0                   |
|---------------------|---------------------|---------------------|---------------------|---------------------|--------------------|--------------------|---------------------|
| VLDx                |                     |                     |                     | PORON               | SVSON              | SVSOP              | SVSFG               |
| rw-0 <sup>(1)</sup> | rw-0 <sup>(1)</sup> | rw-0 <sup>(1)</sup> | rw-0 <sup>(1)</sup> | rw-0 <sup>(1)</sup> | r-0 <sup>(1)</sup> | r-0 <sup>(1)</sup> | rw-0 <sup>(1)</sup> |

Table 9-2. SVSCTL Register Field Descriptions

| Bit | Field | Type | Reset             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----|-------|------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7-4 | VLDx  | R/W  | 0h <sup>(1)</sup> | <p>Voltage level detect. These bits turn on the SVS and select the nominal SVS threshold voltage level. See the device-specific data sheet for parameters.</p> <p>0000b = SVS is off</p> <p>0001b = 1.9 V</p> <p>0010b = 2.1 V</p> <p>0011b = 2.2 V</p> <p>0100b = 2.3 V</p> <p>0101b = 2.4 V</p> <p>0110b = 2.5 V</p> <p>0111b = 2.65 V</p> <p>1000b = 2.8 V</p> <p>1001b = 2.9 V</p> <p>1010b = 3.05 V</p> <p>1011b = 3.2 V</p> <p>1100b = 3.35 V</p> <p>1101b = 3.5 V</p> <p>1110b = 3.7 V</p> <p>1111b = Compares external input voltage SVSIN to 1.25 V</p> |

## SVS Operating Range

**Figure 9-3. SVSCTL Register**

| 7                   | 6                   | 5                   | 4                   | 3                   | 2                  | 1                  | 0                   |
|---------------------|---------------------|---------------------|---------------------|---------------------|--------------------|--------------------|---------------------|
| VLDx                |                     |                     |                     | PORON               | SVSON              | SVSOP              | SVSFG               |
| rw-0 <sup>(1)</sup> | rw-0 <sup>(1)</sup> | rw-0 <sup>(1)</sup> | rw-0 <sup>(1)</sup> | rw-0 <sup>(1)</sup> | r-0 <sup>(1)</sup> | r-0 <sup>(1)</sup> | rw-0 <sup>(1)</sup> |

**Table 9-2. SVSCTL Register Field Descriptions**

| Bit | Field | Type | Reset             | Description                                                                                                                                                                                                                         |
|-----|-------|------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3   | PORON | RW   | 0h <sup>(1)</sup> | POR on. This bit enables the SVSFG flag to cause a POR device reset.<br>0b = SVSFG does not cause a POR<br>1b = SVSFG causes a POR                                                                                                  |
| 2   | SVSON | R    | 0h <sup>(1)</sup> | SVS on. This bit reports the status of SVS operation. This bit DOES NOT turn on the SVS. The SVS is turned on by setting VLDx > 0.<br>0b = SVS is off<br>1b = SVS is on                                                             |
| 1   | SVSOP | R    | 0h <sup>(1)</sup> | SVS output. This bit reflects the output value of the SVS comparator.<br>0b = SVS comparator output is low<br>1b = SVS comparator output is high                                                                                    |
| 0   | SVSFG | RW   | 0h <sup>(1)</sup> | SVS flag. This bit indicates a low-voltage condition. SVSFG remains set after a low-voltage condition until reset by software.<br>0b = No low-voltage condition occurred<br>1b = A low-voltage condition is present or has occurred |

## Watchdog Timer+ (WDT+)

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The primary function of the WDT+ module is to perform a controlled system restart after a software problem occurs.

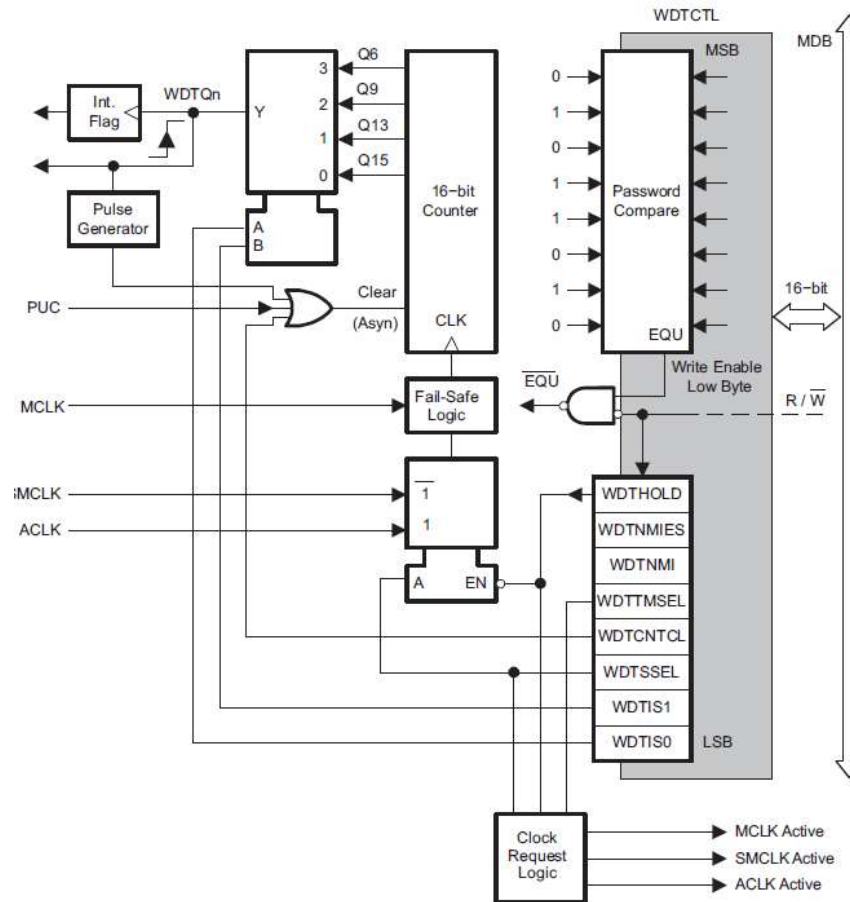
If the selected time interval expires, a system reset is generated.

If the watchdog function is not needed in an application, the module can be configured as an **interval timer** and can **generate interrupts at selected time intervals**.

Features of the watchdog timer+ module include:

- Four software-selectable time intervals
- Watchdog mode
- Interval mode
- Access to WDT+ control register is password protected
- Control of RST/NMI pin function
- Selectable clock source
- Can be stopped to conserve power
- Clock fail-safe feature

## Watchdog Timer+ (WDT+)



## Watchdog Timer+ (WDT+)

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### Watchdog Timer+ Operation

The WDT+ module can be configured as either a watchdog or interval timer with the WDTCTL register.

The WDTCTL register also contains control bits to configure the RST/NMI pin. WDTCTL is a 16-bit, **password protected**, read/write register.

Any read or write access must use word instructions and write accesses must include the write password **0x5Ah** in the upper byte.

Any write to WDTCTL with any value other than 05Ah in the upper byte is a security key violation and triggers a PUC system reset regardless of timer mode.

Any read of WDTCTL reads 069h in the upper byte.

The WDT+ counter clock should be slower or equal than the system (MCLK) frequency.

## Watchdog Timer+ (WDT+)

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### Watchdog Timer+ Counter

The watchdog timer+ counter (WDTCNT) is a 16-bit up-counter that is **not directly accessible** by software.

The WDTCNT is controlled and time intervals selected through the watchdog timer+ control register WDTCTL.

The WDTCNT can be sourced from ACLK or SMCLK.  
The clock source is selected with the WDTSSSEL bit.

## Watchdog Timer+ (WDT+)

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### Watchdog Mode

After a PUC condition, the WDT+ module is configured in the watchdog mode with an initial **32768 cycle reset interval** using the DCOCLK.

The user must setup, halt, or clear the WDT+ prior to the expiration of the initial reset interval or another PUC will be generated.

When the WDT+ is configured to operate in watchdog mode, either writing to WDTCTL with an incorrect password, or expiration of the selected time interval triggers a PUC.

A PUC resets the WDT+ to its default condition and configures the RST/NMI pin to reset mode..



## Watchdog Timer+ (WDT+)

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### Interval Timer Mode

Setting the WDTTMSEL bit to 1 selects the interval timer mode. This mode can be used to provide periodic interrupts.

In interval timer mode, the **WDTIFG flag is set** at the expiration of the selected time interval.

A PUC is **not generated** in interval timer mode at expiration of the selected timer interval and the WDTIFG enable bit WDTIE remains unchanged.

When the WDTIE bit and the GIE bit are set, the WDTIFG flag requests an interrupt.

The **WDTIFG** interrupt flag is **automatically reset** when its interrupt request is serviced, or may be reset by software. The interrupt vector address in interval timer mode is different from that in watchdog mode

## Watchdog Timer+ (WDT+)

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When using the WDT+ in the watchdog mode, the WDTIFG flag sources a **reset vector interrupt**

When using the WDT+ in interval timer mode, the WDTIFG flag is set after the selected time interval and requests a **WDT+ interval timer interrupt** if the **WDTIE** and the **GIE** bits are set

### Watchdog Timer+ Clock Fail-Safe Operation

The WDT+ module provides a fail-safe clocking feature assuring the clock to the WDT+ cannot be disabled while in watchdog mode.

This means the low-power modes may be affected by the choice for the WDT+ clock.

For example, if ACLK is the WDT+ clock source, **LPM4** will not be available, because the WDT+ will prevent ACLK from being disabled.

## Watchdog Timer+ (WDT+)

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If ACLK or SMCLK fail while sourcing the WDT+, the WDT+ clock source is automatically switched to MCLK.

In this case, if MCLK is sourced from a crystal, and the crystal has failed, the fail-safe feature will activate the DCO and use it as the source for MCLK.

When the WDT+ module is used **in interval timer mode**, there is **no fail-safe feature** for the clock source.

## Watchdog Timer+ (WDT+)

---

If ACLK or SMCLK fail while sourcing the WDT+, the WDT+ clock source is automatically switched to MCLK.

In this case, if MCLK is sourced from a crystal, and the crystal has failed, the fail-safe feature will activate the DCO and use it as the source for MCLK.

When the WDT+ module is used **in interval timer mode**, there is **no fail-safe feature** for the clock source.

### Operation in Low-Power Modes

The MSP430 devices have several low-power modes.

Different clock signals are available in different low-power modes.

The requirements of the user's application and the type of clocking used determine how the WDT+ should be configured.

## Watchdog Timer+ (WDT+)

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For example, the WDT+ should not be configured in watchdog mode with SMCLK as its clock source if the user wants to use low-power mode 3 because the WDT+ will keep SMCLK enabled for its clock source, increasing the current consumption of LPM3.

When the watchdog timer+ is not required, the WDT HOLD bit can be used to hold the WDT CNT, reducing power consumption

## Watchdog Timer+ (WDT+)

### Software Examples

Any write operation to WDTCTL must be a **word operation with 05Ah (WDTPW)** in the upper byte:

```
; Periodically clear an active watchdog
```

```
MOV #WDTPW+WDTCNTCL,&WDTCTL
```

```
;
```

```
; Change watchdog timer+ interval
```

```
MOV #WDTPW+WDTCNTL+WDTSSSEL,&WDTCTL
```

```
;
```

```
; Stop the watchdog
```

```
MOV #WDTPW+WDTHOLD,&WDTCTL
```

```
;
```

```
; Change WDT+ to interval timer mode, clock/8192 interval
```

```
MOV #WDTPW+WDTCNTCL+WDTTMSSEL+WDTIS0,&WDTCTL
```

**Table 10-1. Watchdog Timer+ Registers**

| Address | Acronym | Register Name           | Type       | Reset          | Section                        |
|---------|---------|-------------------------|------------|----------------|--------------------------------|
| 120h    | WDTCTL  | Watchdog timer+ control | Read/write | 6900h with PUC | <a href="#">Section 10.3.1</a> |
| 0h      | IE1     | SFR interrupt enable 1  | Read/write | 00h with PUC   | <a href="#">Section 10.3.2</a> |
| 2h      | IFG1    | SFR interrupt flag 1    | Read/write | 00h with PUC   | <a href="#">Section 10.3.3</a> |

## Watchdog Timer+ (WDT+)

Figure 10-2. WDTCTL Register

| 15      | 14       | 13     | 12        | 11       | 10       | 9      | 8    |
|---------|----------|--------|-----------|----------|----------|--------|------|
| WDTPW   |          |        |           |          |          |        |      |
| rw-0    | rw-1     | rw-1   | rw-0      | rw-1     | rw-0     | rw-0   | rw-1 |
| 7       | 6        | 5      | 4         | 3        | 2        | 1      | 0    |
| WDTHOLD | WDTNMIES | WDTNMI | WDTTMSSEL | WDTCNTCL | WDTSSSEL | WDTISx |      |
| rw-0    | rw-0     | rw-0   | rw-0      | r0(w)    | rw-0     | rw-0   | rw-0 |

| Bit  | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                     |
|------|----------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-8 | WDTPW    | R/W  | 69h   | Watchdog timer+ password. Always reads as 069h. Must be written as 05Ah. Writing any other value generates a PUC.                                                                                                                                                               |
| 7    | WDTHOLD  | R/W  | 0h    | Watchdog timer+ hold. This bit stops the watchdog timer+. Setting WDTHOLD = 1 when the WDT+ is not in use conserves power.<br>0b = Watchdog timer+ is not stopped<br>1b = Watchdog timer+ is stopped                                                                            |
| 6    | WDTNMIES | R/W  | 0h    | Watchdog timer+ NMI edge select. This bit selects the interrupt edge for the NMI interrupt when WDTNMI = 1. Modifying this bit can trigger an NMI. Modify this bit when WDTIE = 0 to avoid triggering an accidental NMI.<br>0b = NMI on rising edge<br>1b = NMI on falling edge |
| 5    | WDTNMI   | R/W  | 0h    | Watchdog timer+ NMI select. This bit selects the function for the RST/NMI pin.<br>0b = Reset function<br>1b = NMI function                                                                                                                                                      |

## Watchdog Timer+ (WDT+)

| Bit | Field     | Type | Reset | Description                                                                                                                                                                                                                                                                 |
|-----|-----------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4   | WDTTMSSEL | R/W  | 0h    | Watchdog timer+ mode select<br>0b = Watchdog mode<br>1b = Interval timer mode                                                                                                                                                                                               |
| 3   | WDCNTCL   | R/W  | 0h    | Watchdog timer+ counter clear. Setting WDCNTCL = 1 clears the count value to 0000h. WDCNTCL is automatically reset.<br>0b = No action<br>1b = WDCNT = 0000h                                                                                                                 |
| 2   | WDTSSEL   | R/W  | 0h    | Watchdog timer+ clock source select<br>0b = SMCLK<br>1b = ACLK                                                                                                                                                                                                              |
| 1-0 | WDTISx    | R/W  | 0h    | Watchdog timer+ interval select. These bits select the watchdog timer+ interval to set the WDTIFG flag or generate a PUC.<br>00b = Watchdog clock source /32768<br>01b = Watchdog clock source /8192<br>10b = Watchdog clock source /512<br>11b = Watchdog clock source /64 |



## Watchdog Timer+ (WDT+)

**Figure 10-3. IE1 Register**

| 7    | 6 | 5 | 4     | 3    | 2 | 1 | 0     |
|------|---|---|-------|------|---|---|-------|
|      |   |   | NMIIE |      |   |   | WDTIE |
| rw-0 |   |   |       | rw-0 |   |   |       |

**Table 10-3. IE1 Register Field Descriptions**

| Bit | Field | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                     |
|-----|-------|------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7-5 |       |      |       | These bits may be used by other modules. See device-specific data sheet.                                                                                                                                                                                                                                                                                                                        |
| 4   | NMIIE | R/W  | 0h    | NMI interrupt enable. This bit enables the NMI interrupt. Because other bits in IE1 may be used for other modules, it is recommended to set or clear this bit using BIS.B or BIC.B instructions, rather than MOV.B or CLR.B instructions.<br>0b = Interrupt not enabled<br>1b = Interrupt enabled                                                                                               |
| 3-1 |       |      |       | These bits may be used by other modules. See device-specific data sheet.                                                                                                                                                                                                                                                                                                                        |
| 0   | WDTIE | R/W  | 0h    | Watchdog timer+ interrupt enable. This bit enables the WDTIFG interrupt for interval timer mode. It is not necessary to set this bit for watchdog mode. Because other bits in IE1 may be used for other modules, it is recommended to set or clear this bit using BIS.B or BIC.B instructions, rather than MOV.B or CLR.B instructions.<br>0b = Interrupt not enabled<br>1b = Interrupt enabled |

## Watchdog Timer+ (WDT+)

**Figure 10-4. IFG1 Register**

| 7 | 6 | 5 | 4      | 3 | 2 | 1 | 0      |
|---|---|---|--------|---|---|---|--------|
|   |   |   | NMIIFG |   |   |   | WDTIFG |
|   |   |   | rw-0   |   |   |   | rw-(0) |

**Table 10-4. IFG1 Register Field Descriptions**

| Bit | Field  | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----|--------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7-5 |        |      |       | These bits may be used by other modules. See device-specific data sheet.                                                                                                                                                                                                                                                                                                                                                                 |
| 4   | NMIIFG | R/W  | 0h    | NMI interrupt flag. NMIIFG must be reset by software. Because other bits in IFG1 may be used for other modules, it is recommended to clear NMIIFG by using BIS.B or BIC.B instructions, rather than MOV.B or CLR.B instructions.<br>0b = No interrupt pending<br>1b = Interrupt pending                                                                                                                                                  |
| 3-1 |        |      |       | These bits may be used by other modules. See device-specific data sheet.                                                                                                                                                                                                                                                                                                                                                                 |
| 0   | WDTIFG | R/W  | 0h    | Watchdog timer+ interrupt flag. In watchdog mode, WDTIFG remains set until reset by software. In interval mode, WDTIFG is reset automatically by servicing the interrupt, or can be reset by software. Because other bits in IFG1 may be used for other modules, it is recommended to clear WDTIFG by using BIS.B or BIC.B instructions, rather than MOV.B or CLR.B instructions.<br>0b = No interrupt pending<br>1b = Interrupt pending |



## THANK YOU

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